

5 Photosynthesis consists of the light-dependent reactions and the light-independent reactions.

(a) In the light-dependent reactions, light energy is converted into energy stored in ATP.

(i) Explain why light energy is converted into energy stored in ATP.

(2)

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(ii) An equation can summarise the production and breakdown of ATP.

Complete this equation, by writing the names of the substrates and the type of reaction on the three dotted lines provided.

(2)



(iii) Explain the role of light energy in the light-dependent reactions.

(3)

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(b) The light-independent reactions use products of the light-dependent reactions to produce simple sugars.

(i) Which row of the table shows the products of the light-dependent reactions that are used in the light-independent reactions?

(1)

	ATP produced by	NADP
☐ A	cyclic photophosphorylation	oxidised
☐ B	cyclic photophosphorylation	reduced
☐ C	non-cyclic photophosphorylation	oxidised
☐ D	non-cyclic photophosphorylation	reduced

(ii) Simple sugars have the formula $C_nH_{2n}O_n$.

Name the inorganic molecule from which each element in a simple sugar originated.

(2)

C

H

O

(iii) Simple sugars are used in the synthesis of new biological molecules.

Which row of the table shows the inorganic ions that are needed to synthesise these new biological molecules?

(3)

New biological molecule	Nitrates	Phosphates	Both nitrates and phosphates	Neither nitrates nor phosphates
protein	☐	☐	☐	☐
RNA	☐	☐	☐	☐
triglyceride	☐	☐	☐	☐

(Total for Question 5 = 13 marks)